

Lab 3 – Quarto Website

In this lab, you will be creating a website hosted on GitHub pages that contains a portfolio of the work you've done in this class.

Setup

1. We will begin by setting up a Quarto website. Instructions are below and can also be found on this page: <https://quarto.org/docs/websites/>.

In RStudio, do the following to create a Quarto Website Project:

- a. Click on the Project option in the top right of the program.
- b. Select New Project....
- c. Select New Directory.
- d. Select Quarto Website.
- e. Name the directory that will store your website files and put it in your Math 3190 folder on your computer. You can keep the default options checked and unchecked and select Create Project.

This will create a folder with `_quarto.yml`, `about.qmd`, `index.qmd`, `.Rproj`, and `styles.css` files.

In Positron, do the following to create a Quarto Website folder:

- a. Open the Command Palette by typing Command+Shift+P on a Mac or Control+Shift+P on Windows.
- b. In the Command Palette, type Quarto: Create Project and select that option.
- c. Click on Website Project.
- d. Navigate to your Math 3190 folder on your computer and click on Choose Parent Directory.
- e. Name the directory that will store your website files.

This will create a folder with `_quarto.yml`, `about.qmd`, `index.qmd`, and `styles.css` files.

2. Now we want to link the website folder with GitHub:
 - a. Create a new GitHub repository that is named something you'd like in your website URL. The URL will be `https://username.github.io/repo_name`. You can add a description, but don't change the other options.
 - b. On your computer, open a Terminal and navigate to the folder you set up earlier with your website files in it. Then follow the **"...or create a new repository on the command line"** options to set up and link the folder with your GitHub repo.

Building Website

3. Now it's time to start building the website.
 - a. Make sure you are in your website project in RStudio or Positron. You will know because the name of project (or folder in Positron) will appear in the top-left of the window.
 - b. Open the `_quarto.yml` file (it might already be open by default) in your IDE.
 - c. The first two lines should currently be

```
project:
type: website
```

Modify this so the first three lines are

project:

type: website

output-dir: docs

Note that the **type** and **output-dir** lines are both indented.

- d. In your website folder on your computer, create a Homework folder and copy (not move) your homework Quarto (.qmd) files and any helper files like data or images or cache folders to that folder.
- e. Once you've done that, we can add those Quarto as pages to your website. Under the - **about.qmd** line in the **_quarto.yml** file, you can add lines to add pages to the site. For example, adding
- **href: Homework/Homework_1.qmd**
text: Homework 1
will add a page with your first homework assignment on it. Of course, make sure the name of the .qmd file matches the one that is actually in the Homework folder.

Deploying Website

4. Now we need to render the website.

In RStudio:

- a. When working in the website project environment, there should be a Build pane near the Environment pane near the top right of the window. Click that.
- b. Click Render Website. This will render all of your homework assignments, so it could take a little while. If it takes too long, or gives problems when rendering it, you could put the .html files instead of .qmd files in Step 3e above, but it doesn't work as well as using the .qmd file.
- c. Once it renders, this should give you a preview of your website in the Viewer pane. You can click the Show in new window button (with an arrow pointing up and right) to open it in a new window so you can navigate the website. This should also add a docs/ folder with files in it among other things like a .gitignore file. You can open the .gitignore and add a few lines like .DS_Store (if on a Mac) and .gitignore to the .gitignore.

In Positron:

- a. When working in the website project folder, open the command palette (Command + Shift + P or Control + Shift + P) and type Quarto: Render Project. This will render all of your homework assignments, so it could take a little while. If it takes too long, or gives problems when rendering it, you could put the .html files instead of .qmd files in Step 3e above, but it doesn't work as well as using the .qmd file.
 - b. You can view the website in Positron's viewer by opening the index.qmd file and clicking Preview or typing Command + Shift + K (Mac) or Control + Shift + K (Windows).
5. Add, commit, and push all the contents of this website folder to GitHub.
 6. On GitHub, go to your website directory and do the following:
 - a. Click on Settings.
 - b. Go down to Pages on the left side.
 - c. Under Branch, select None and change it to main.

- d. Change /(root) to /docs.
- e. Click Save.
7. After a moment or two, the page should deploy. When back on your main website repo page, on the right side it should have a **Deployments** section with github-pages underneath. Click on the github-pages link.
8. Then click on the URL under git-hub pages to see your website.

Now when you make changes to the files and push them to GitHub, it will automatically deploy the updated website (it may take a few minutes after each push to update it).

There is so much customization you can do with your website. You can check out one I made here: <https://rbrown53.github.io/Math3190Website/> (not that it is particularly impressive). The repository for it with the `_quarto.yml` file can be found here: <https://github.com/rbrown53/Math3190Website>. Feel free to look at that file and experiment with your website. There is also code on there to include pdf files if you have any you need to include. Of course, my website only has three homework assignments on it, yours should have all of them. Mine also has some Lab assignments that we did not do this semester.

There is also lots of helpful info on the Quarto help pages: <https://quarto.org/docs/websites/website-navigation.html>.

Customizing Website

9. To finish this lab, add each of your homework assignments to the website.
10. Customize your website so you move away from the default theme, title, and navbar navigation. Do something to make it more personal.
11. Add information to the Home page (the `Index.qmd` file) about what this website is for and what it contains.
12. Remove the **about.qmd** line from the `_quarto.yml` since it is a bit redundant for this website.

Closing Project or Folder

13. In RStudio:

When you are finished, you can close the project by clicking on the project name in the top right of the window and going down to Close Project. You can reopen the website project by opening the `.Rproj` file in your website folder. If you don't close this, it may automatically open the next time you want to use RStudio, which you likely don't want.

In Positron:

You can close the folder by clicking on the folder name in the top right of the window and going down to Close Folder. You can reopen the website folder by clicking that folder icon in the top right again and navigating to your website folder. If you don't close this, it may automatically open the next time you want to use Positron, which you likely don't want.